



Weighted substring

Every character has a weight. The weight of an English uppercase alphabet A-Z is given below. $A=1$ $B=2*A+A$ $C=3*B+B$ $D=4*C+C$. . . $Y=25*X+X$ $Z=26*Y+Y$ The weight of any string made up of these characters is the summation of weights of each character. Given a total string weight, determine shortest string of that given weight. If there is more than one solution, return the lexicographically smallest of them. For example, given weight = 25, and the weights of the first few characters of the alphabet are $A=1$, $B=3$, $C=12$, $D=60$ it is certain that no letter larger than C is required. Some of the strings with a total weight equal to the largest are ABBBBC, ACC, and AAAAAAABBBBBB. The shortest of these is ACC. While any permutation of these characters will have the same weight, this is the lexicographically smallest of them. Function Description Complete the function smallestString in the editor below.

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The function must return the shortest string of the target weight. If there are multiple answers, return the lexicographically smallest of them.

smallestString has the following parameter(s): weight: a long integer that denotes the target weight.

Input Format

Input consists of a string s .

Output Format

Output should be in the form of integer.

Constraints

$1 \leq \text{weight} \leq 1016$

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**Sample
Input:**

20

**Sample
Output:**

AABBC



THANK YOU